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Practical No. 6 Implementation of Complex Queries using group by, joins

Name of Expt. Implementation of Complex Queries using group by, order by, having clause , joins

Theory

Group by clause –

- The GROUP BY clause is a SQL command that is used to **group rows that have the same values.**
- The queries that contain the GROUP BY clause are called grouped queries and only return a single row for every grouped item.
- The GROUP BY clause is used in the SELECT statement

HAVING clause –

- The HAVING clause is used in the SELECT statement to specify filter conditions for a group of rows or aggregates.
- The HAVING clause is often used with the GROUP BY clause to filter groups based on a specified condition. If the GROUP BY clause is omitted, the HAVING clause behaves like the WHERE clause.
- The HAVING clause applies a filter condition to each group of rows, while the WHERE clause applies the filter condition to each individual row.

ORDER BY clause - The MySQL ORDER BY clause is used to sort the records in your result set

Group by clause

(i)Find grouped salaries of employees. (group by clause)

SQL> select salary from emp group by salary;

SALARY

- 9000

10000

8000

6000

7000

(ii).Find salaries of employee in Ascending Order. (order by clause) SQL> select ename,salary from emp order by salary;

ENAME	SALARY
rohan	6000

alex 7000
shane 8000
abhi 8000
tiger 8000
anu 9000
scott 10000

7 rows selected.
(vii) Find salaries of employee in Descending Order.
SQL> select ename, salary from emp order by salary desc; ENAME SALARY
---------- ----------
scott 10000
anu 9000
shane 8000
abhi 8000
tiger 8000
alex 7000
rohan 6000

7 rows selected.

(xii) Having Clause.
SQL> select ename,salary from emp where age<29 group by ename,salary having salary<10000; ENAME SALARY
---------- ----------
alex 7000
anu 9000

MySQL JOINS

MySQL JOINS are used with SELECT statement. It is used to retrieve data from multiple tables. It is performed whenever you need to fetch records from two or more tables.
There are three types of MySQL joins:

- MySQL CROSS JOIN
- MySQL INNER JOIN (or sometimes called simple join)
- MySQL LEFT OUTER JOIN (or sometimes called LEFT JOIN)
- MySQL RIGHT OUTER JOIN (or sometimes called RIGHT JOIN)

Cross JOIN

Cross JOIN is a simplest form of JOINs which matches each row from one database table to all rows of

another.
In other words it gives us combinations of each row of first table with all records in second table. Syntax –

```
mysql> select * from first_table_name cross join
another_table_name; Ex-
mysql> select * from stu;
+___+____+
| id | name |
+___+____+
| 123 | s    |
| 2   | sdf |
| 6   | df   |
+___+____+
3 rows in set (0.00 sec)

mysql> select * from stu_m;
+____+____+
| id | sgpi |
+____+____+
|    1 | 8.70 |
|    2 | 6.70 |
|    4 | 7.70 |
+____+____+
3 rows in set (0.00 sec)

mysql> select * from stu cross join stu_m;
+___+____+____+____+
| id | name | id | sgpi |

| 123 | s    |    1 | 8.70 |
| 2 | sdf |     1 | 8.70 |
| 6 | df   |     1 | 8.70 |
| 123 | s    |    2 | 6.70 |
| 2 | sdf |     2 | 6.70 |
| 6 | df   |     2 | 6.70 |
| 123 | s    |    4 | 7.70 |
| 2 | sdf |     4 | 7.70 |
| 6 | df   |     4 | 7.70 |
+___+____+____+__
+___+____+____+____+
```

9 rows in set (0.00 sec)

mysql> select * from stu cross join stu_m;

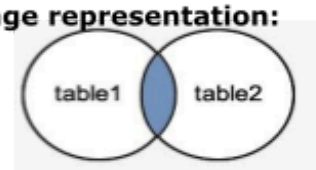
MySQL Inner JOIN (Simple Join/ cross join (without condition))

The MySQL INNER JOIN is used to return all rows from multiple tables where the join condition is satisfied. It is the most common type of join.

Syntax:

```
SELECT columns
FROM table1
INNER JOIN table2
ON table1.column = table2.column;
```

Image representation:




```
mysql>
select stu.id, stu.name, stu_m.id from stu inner join stu_m on stu.id=stu_m.id;
+      +      +      +
| id | name | id |
+      +      +      +
| 2 | sdf |      2 |
+      +      +      +
1 row in set (0.04 sec)
```

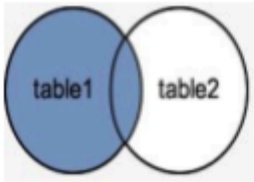
MySQL Left Outer Join

The LEFT OUTER JOIN returns all rows from the left hand table specified in the ON condition and only those rows from the other table where the join condition is fulfilled.

Syntax:

```
SELECT columns
FROM table1
LEFT [OUTER] JOIN table2
ON table1.column = table2.column;
```

Image representation:




```
mysql> select stu.id, stu.name, stu_m.id from stu left join stu_m on stu.id=stu_m.id;
+___+____+____+
| id | name | id |
+___+____+____+
| 2   | sdf |     2 |
| 123 | s    | NULL |
| 6 | df   | NULL |
+___+____+____+
3 rows in set (0.00 sec)
```

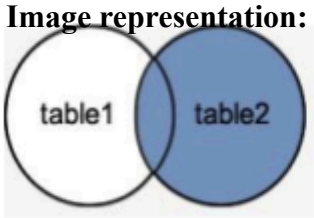
MySQL Right Outer Join

The MySQL Right Outer Join returns all rows from the RIGHT-hand table specified in the ON condition and only those rows from the other table where he join condition is fulfilled.

Syntax:

SELECT columns
FROM table1
RIGHT [OUTER] JOIN table2
ON table1.**column** = table2.**column**;

Image representation:



Ex

```
mysql> select stu.id, stu.name, stu_m.id from stu right join stu_m on stu.id=stu_m.id;
+------+------+------+
| id | name | id   |
+------+------+------+
| 2    | sdf |     2 |
| NULL | NULL |    1 |
| NULL | NULL |    4 |
```

+------+------+------+
3 rows in set (0.00 sec)

CONCLUSION: Student are able to implement given queries using group by, order by, having clause , can join tables and create virtual tables.

VIVA QUESTIONS

1. What is Join?
2. Mention the different types of Join operations.
3. What is the difference between Inner Join & Outer Join.
4. What are the prerequisites for any two tables to apply a Join?
5. Give practical examples where we use joins?